

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250281706

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

WADA; Hiroaki

SILENCER FOR CPAP DEVICE AND CPAP DEVICE

Abstract

A silencer for a CPAP device comprises a circulation channel through which air can be circulated and a first silencing chamber to which pressure fluctuations of the air in the circulation channel can be transmitted. The circulation channel includes a first flow channel, a second flow channel, and a third flow channel that extend along a first axis. The circulation channel further includes a first connecting flow channel that connects an end of the first flow channel on a first positive direction side and an end of the second flow channel on the first positive direction side, and a second connecting flow channel that connects an end of the second flow channel on a first negative direction side and an end of the third flow channel on the first negative direction side.

Inventors: WADA; Hiroaki (Nagaokakyo-shi, JP)

Applicant: Murata Manufacturing Co., Ltd. (Nagaokakyo-shi, JP)

Family ID: 91323600

Appl. No.: 19/217116

Filed: May 23, 2025

Foreign Application Priority Data

JP 2022-190489

Nov. 29, 2022

Related U.S. Application Data

parent WO continuation PCT/JP2023/037878 20231019 PENDING child US 19217116

Publication Classification

Int. Cl.: A61M16/00 (20060101)

U.S. Cl.:

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This is a continuation of International Application No. PCT/JP2023/037878 filed on Oct. 19, 2023 which claims priority from Japanese Patent Application No. 2022-190489 filed on Nov. 29, 2022. The contents of these applications are incorporated herein by reference in their entireties.

BACKGROUND OF THE DISCLOSURE

Field of the Disclosure

[0002] The present disclosure relates to a silencer for a continuous positive airway pressure (CPAP) device and a CPAP device.

Description of the Related Art

[0003] International Publication No. WO2019/189126A1 describes a CPAP device that includes a main unit and a sub-unit. The main unit incorporates a blower. The sub-unit incorporates a silencer. The silencer includes a flow passage of air for the blower, and a resonance tube branched from the flow passage. The resonance tube is a side branch type resonator.

BRIEF SUMMARY OF THE DISCLOSURE

[0004] In a CPAP device, such as that of International Publication No. WO2019/189126A1, the amount of air sent from the blower needs to be changed in accordance with breathing of a user. Therefore, the CPAP device changes a rotational speed or the like of a fan of the blower in accordance with the breathing of the user. Accordingly, in the CPAP device, noise may occur in various frequency bands. The CPAP device of International Publication No. WO2019/189126A1 has a noise reduction effect on a specific frequency that corresponds to the shape of the resonance tube. However, the noise reduction effect of the CPAP device of International Publication No. WO2019/189126A1 may be limited in a frequency band outside the specific frequency.

[0005] A silencer for a CPAP device is provided to solve the above problem. In one general aspect, the silencer includes a flow passage through which air flows in accordance with operation of a blower, and a noise cancellation chamber that allows for transmission of pressure fluctuations of the air in the flow passage. The flow passage includes a first flow passage, a second flow passage, a third flow passage, a first connection passage, and a second connection passage. The first flow passage extends along a first axis. The second flow passage extends along the first axis and is arranged next to the first flow passage in a direction orthogonal to the first axis. The third flow passage extends along the first axis and is arranged next to the second flow passage in the direction orthogonal to the first axis. The first connection passage connects an end of the first flow passage in a positive direction and an end of the second flow passage in the positive direction. The positive direction is one of two directions parallel to the first axis. The second connection passage connects an end of the second flow passage in a negative direction and an end of the third flow passage in the negative direction. The negative direction is another one of the two directions parallel to the first axis and is opposite to the positive direction. A maximum dimension of the first connection passage in a direction in which the second flow passage is arranged next to the third flow passage differs from a maximum dimension of the second connection passage in the direction in which the second flow passage is arranged next to the third flow passage. The noise cancellation chamber is located between the second flow passage and the third flow passage in the direction in which the second flow passage is arranged next to the third flow passage. The noise cancellation chamber is connected to the second connection passage through an opening included in a wall that separates the noise cancellation chamber and the second connection passage.

[0006] In another general aspect, a CPAP device includes a blower that blows air, and a silencer.

The silencer includes a flow passage through which air flows in accordance with operation of the blower, and a noise cancellation chamber that allows for transmission of pressure fluctuations of the air in the flow passage. The flow passage includes a first flow passage, a second flow passage, a third flow passage, a first connection passage, and a second connection passage. The first flow passage extends along a first axis. The second flow passage extends along the first axis and is arranged next to the first flow passage in a direction orthogonal to the first axis. The third flow passage extends along the first axis and is arranged next to the second flow passage in the direction orthogonal to the first axis. The first connection passage connects an end of the first flow passage in a positive direction and an end of the second flow passage in the positive direction. The positive direction is one of two directions parallel to the first axis. The second connection passage connects an end of the second flow passage in a negative direction and an end of the third flow passage in the negative direction. The negative direction is another one of the two directions parallel to the first axis and is opposite to the positive direction. A maximum dimension of the first connection passage in a direction in which the second flow passage is arranged next to the third flow passage differs from a maximum dimension of the second connection passage in the direction in which the second flow passage is arranged next to the third flow passage. The noise cancellation chamber is located between the second flow passage and the third flow passage in the direction in which the second flow passage is arranged next to the third flow passage. The noise cancellation chamber is connected to the second connection passage through an opening included in a wall that separates the noise cancellation chamber and the second connection passage.

[0007] With the above structure, the maximum dimension of the first connection passage in the direction in which the second flow passage is arranged next to the third flow passage differs from the maximum dimension of the second connection passage in the direction in which the second flow passage is arranged next to the third flow passage. Such a difference in the dimension of the connection passages allows for cancellation of noise in different frequency bands in the connection passages. Consequently, noise in various frequency bands is reduced. Furthermore, the silencer of the CPAP device includes the noise cancellation chamber, and the opening continuous with the noise cancellation chamber is arranged at a position where the flow passage is bent. This facilitates propagation of pressure fluctuations of air, or noise, into the noise cancellation chamber.

Consequently, noise is effectively cancelled in the noise cancellation chamber.

[0008] Noise in various frequency bands may be reduced or cancelled.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0009] FIG. 1 is a schematic diagram showing a CPAP device in use.

[0010] FIG. 2 is an exploded perspective view of the CPAP device.

[0011] FIG. 3 is a schematic diagram showing the configuration of the CPAP device.

[0012] FIG. 4 is a diagram of the CPAP device shown in FIG. 2, as seen through a bottom surface.

DETAILED DESCRIPTION OF THE DISCLOSURE

<Embodiment of Silencer for CPAP Device>

[0013] An embodiment of a silencer for a continuous positive airway pressure (CPAP) device will now be described with reference to the drawings. Depiction of elements in the drawings may be exaggerated for clarity, illustration, or convenience. The elements may not be drawn to scale, and the relative size and proportions of the elements may differ between the drawings.

(Structure of CPAP Device)

[0014] The overall configuration of the CPAP device **10** will be described. As shown in FIGS. 1 and 2, a CPAP device **10** includes a main unit MU and a sub-unit SU. In a used state in which the main unit MU is attached to the sub-unit SU, that is, in a state in which the CPAP device **10** is in

use, the CPAP device **10** has the form of a box as a whole.

[0015] As shown in FIG. 2, the main unit MU of the CPAP device **10** is substantially box-shaped. As shown in FIG. 3, the main unit MU includes a blower **20**, an operating portion **21**, a controller **100**, and a power supply **101**.

[0016] The blower **20** accommodates a fan (not shown) that blows air. The operating portion **21** is disposed on an upper surface of the main unit MU. The operating portion **21** is used to control ON/OFF states or the like of the blower **20** via the controller **100**.

[0017] The controller **100** receives a signal corresponding to an operation performed on the operating portion **21**. Based on the signal received from the operating portion **21**, the controller **100** outputs a control signal to the blower **20**. The controller **100** controls the ON/OFF states, rotational speed, or the like of the blower **20** using the control signal.

[0018] The controller **100** may be circuitry including one or more processors that run computer programs (software) to execute various processes, one or more dedicated hardware circuits such as application-specific integrated circuits (ASICs) that execute at least some of the various processes, or a combination of the above.

[0019] The processor includes a central processing unit (CPU) and memory, such as random-access memory (RAM), read-only memory (ROM), or the like. The memory stores program codes or instructions configured to cause the CPU to execute processes. The memory, which is a computer readable medium, may be any available medium accessible by a versatile or dedicated computer.

[0020] The power supply **101** supplies power to, for example, the controller **100** and the blower **20**. Specifically, the power supply **101** includes a battery. Specifically, the power supply **101** includes a rechargeable battery that can be charged repeatedly.

[0021] The main unit MU includes an upstream passage **61** and a downstream passage **62**. The upstream passage **61** and the downstream passage **62** are arranged inside the main unit MU. The upstream passage **61** and the downstream passage **62** are connected to the blower **20**. The blower **20** draws air from the upstream passage **61** and blows the air to the downstream passage **62**.

[0022] As shown in FIG. 2, the sub-unit SU of the CPAP device **10** is L-shaped in side view taken in a direction orthogonal to a longitudinal direction. When the main unit MU is attached to the sub-unit SU, the CPAP device **10** becomes substantially box-shaped, as described above. The sub-unit SU is attachable to and detachable from the main unit MU.

[0023] As shown in FIG. 3, the sub-unit SU includes a humidifier **30** and a silencer **40**. That is, the sub-unit SU corresponds to the silencer for a CPAP device. The humidifier **30** is configured to store water. The air blown by the fan of the blower **20** may flow into the humidifier **30**. The silencer **40** is configured to reduce noise generated by operation of the blower **20**. The shape or the like of the silencer **40** will be described in detail later.

[0024] As shown in FIG. 2, an axis extending in the longitudinal direction of the sub-unit SU is referred to as a first axis X. An axis orthogonal to the first axis X is referred to as a second axis Y. In the present embodiment, the second axis Y extends in a transverse direction of the sub-unit SU. An axis orthogonal to the first axis X and the second axis Y is referred to as a third axis Z. One of two directions parallel to the first axis X is referred to as a first positive direction X1, and the other one of the two directions parallel to the first axis X is referred to as a first negative direction X2. The first negative direction X2 is opposite to the first positive direction X1. One of two directions parallel to the second axis Y is referred to as a second positive direction Y1, and the other one of the two directions parallel to the second axis Y is referred to as a second negative direction Y2. The second negative direction Y2 is opposite to the second positive direction Y1. One of two directions parallel to the third axis Z is referred to as a third positive direction Z1, and the other one of the two directions parallel to the third axis Z is referred to as a third negative direction Z2. The third negative direction Z2 is opposite to the third positive direction Z1.

[0025] As shown in FIG. 3, the sub-unit SU includes a first inflow passage **51**, a first outflow passage **52**, a second inflow passage **53**, and a second outflow passage **54**.

[0026] The first inflow passage **51** is arranged inside the sub-unit SU. The first inflow passage **51** has a first end that is open to the outside of the sub-unit SU. The first inflow passage **51** has a second end that is connected to the silencer **40**. The first outflow passage **52** is arranged inside the sub-unit SU. The first outflow passage **52** has a first end that is connected to the silencer **40**. The first outflow passage **52** has a second end that is open to the outside of the sub-unit SU.

[0027] The second inflow passage **53** is arranged inside the sub-unit SU. The second inflow passage **53** has a first end that is open to the outside of the sub-unit SU. The second inflow passage **53** has a second end that is connected to the humidifier **30**. The second outflow passage **54** is arranged inside the sub-unit SU. The second outflow passage **54** has a first end that is connected to the humidifier **30**. The second outflow passage **54** has a second end that is open to the outside of the sub-unit SU.

[0028] As shown in FIG. **1**, the CPAP device **10** includes a hose **91** and a mask **92**. The hose **91** has a first end connected to the second end of the second outflow passage **54**. The hose **91** has a second end connected to the mask **92**. The mask **92** is attached to a user **93** so as to cover the nose or the mouth of the user **93**. Although not illustrated, the first end of the hose **91** may be connected to the downstream passage **62** of the main unit MU.

(Flow Passage of Air in Used State)

[0029] As described above, the sub-unit SU is attachable to the main unit MU. In a state in which the sub-unit SU is attached to the main unit MU, the passages of the main unit MU and the passages of the sub-unit SU are all connected to form a single flow passage. Specifically, as shown in FIG. **3**, the upstream passage **61** of the main unit MU is connected to the second end of the first outflow passage **52** of the sub-unit SU. Further, the downstream passage **62** of the main unit MU is connected to the first end of the second inflow passage **53** of the sub-unit SU. Thus, when the blower **20** is operated, air is drawn from the first inflow passage **51** and flows through the silencer **40**, the first outflow passage **52**, and the upstream passage **61** to the blower **20**. Then, the air is blown by the blower **20** and flows through the downstream passage **62**, the second inflow passage **53**, the second outflow passage **54**, and the hose **91** to the mask **92**.

[0030] Although not illustrated, the CPAP device **10** may also be used in a state in which the sub-unit SU is detached from the main unit MU. In this case, air flows through the upstream passage **61**, the blower **20**, the downstream passage **62**, and the hose **91** to the mask **92**.

(Silencer of CPAP Device)

[0031] The structure of the silencer **40** and its periphery will now be described. As shown in FIG. **4**, the silencer **40** includes a flow passage of air arranged inside the sub-unit SU. In FIG. **4**, the sub-unit SU is seen through a bottom lid **60** that seals a side of the sub-unit SU oriented in the third negative direction Z2. The bottom lid **60** is indicated by the double-dashed lines.

[0032] The above-described first inflow passage **51** extends along the first axis X inside the sub-unit SU. The first end of the first inflow passage **51** is open in a surface of the sub-unit SU that is oriented in the first negative direction X2. The silencer **40** of the CPAP device **10** includes a flow passage **41**. The flow passage **41** is a passage through which air flows in accordance with operation of the blower **20**. The flow passage **41** is defined by walls **44**. An upstream end of the flow passage **41** is connected to the second end of the first inflow passage **51**.

[0033] The flow passage **41** includes a first flow passage **41A**, a second flow passage **41B**, a third flow passage **41C**, a first connection passage **41D**, and a second connection passage **41E**. The first flow passage **41A** extends along the first axis X. An end of the first flow passage **41A** in the first negative direction X2 is connected to the second end of the first inflow passage **51**. The second flow passage **41B** extends along the first axis X. The second flow passage **41B** is arranged next to the first flow passages **41A** in the second positive direction Y1. The second flow passage **41B** and the first flow passage **41A** are arranged side by side with a single wall **44** in between.

[0034] The third flow passage **41C** extends along the first axis X. The third flow passage **41C** is arranged next to the second flow passage **41B** in the second positive direction Y1. The third flow

passage **41C** and the second flow passage **41B** are spaced apart from each other in a direction parallel to the second axis Y.

[0035] The first connection passage **41D** extends along the second axis Y. An end of the first connection passage **41D** in the second negative direction Y2 is connected to an end of the first flow passage **41A** in the first positive direction X1. An end of the first connection passage **41D** in the second positive direction Y1 is connected to an end of the second flow passage **41B** in the first positive direction X1. Thus, the first flow passage **41A**, the first connection passage **41D**, and the second flow passage **41B** form a U-shaped passage as a whole.

[0036] The second connection passage **41E** extends along the second axis Y. An end of the second connection passage **41E** in the second negative direction Y2 is connected to an end of the second flow passage **41B** in the first negative direction X2. An end of the second connection passage **41E** in the second positive direction Y1 is connected to an end of the third flow passage **41C** in the first negative direction X2. Thus, the second flow passage **41B**, the second connection passage **41E**, and the third flow passage **41C** form a U-shaped passage as a whole.

[0037] A maximum dimension of the first connection passage **41D** in a direction parallel to the second axis Y is referred to as a first connection passage length **41DL**. A maximum dimension of the second connection passage **41E** in a direction parallel to the second axis Y is referred to as a second connection passage length **41EL**. The first connection passage length **41DL** differs from the second connection passage length **41EL**. Specifically, the second connection passage length **41EL** is greater than the first connection passage length **41DL** due to the space between the second flow passage **41B** and the third flow passage **41C**.

[0038] The above-described first outflow passage **52** extends along the third axis Z inside the sub-unit SU. The first end of the first outflow passage **52** is connected to an end of the third flow passage **41C** in the first positive direction X1. The second end of the first outflow passage **52** is open in a surface of the sub-unit SU that is oriented in the first negative direction X2. More specifically, the second end of the first outflow passage **52** is open in an inner surface of the L-shape of the substantially L-shaped sub-unit SU.

[0039] When the blower **20** of the CPAP device **10** is operated, air flows through the silencer **40**. Specifically, air flows through the first flow passage **41A**, the first connection passage **41D**, the second flow passage **41B**, the second connection passage **41E**, and the third flow passage **41C** in this order. Therefore, the second connection passage **41E** is located closer to the blower **20** than the first connection passage **41D** is with respect to an airflow direction.

[0040] An imaginary line extending along the center of a passage from one end to the other end serves as a center axis of the passage. If the center axis of the passage and the first axis X form an acute angle that is 0 degrees or greater and less than 45 degrees, it is considered that the passage extends along the first axis X. In the present embodiment, each of the first to third flow passages **41A** to **41C** forms substantially 0 degrees with the first axis X. The same also applies to a relationship between the second axis Y and a flow passage and a relationship between the third axis Z and a flow passage.

[0041] The boundary between the first flow passage **41A** and the first connection passage **41D** and the boundary between the second flow passage **41B** and the first connection passage **41D** are defined as described below. An imaginary plane orthogonal to the first axis X, which is parallel to the center axis of the first flow passage **41A**, is referred to as a first imaginary plane B1. The first imaginary plane B1 is in contact with the wall **44** that separates the first flow passage **41A** and the second flow passage **41B**. Further, the first imaginary plane B1 is located at an end of the wall **44** toward the first connection passage **41D**. The portion extending from the first imaginary plane B1 in the first positive direction X1 corresponds to the first connection passage **41D**. In the same manner, another imaginary plane orthogonal to the first axis X is referred to as a second imaginary plane B2. The second imaginary plane B2 is in contact with the wall **44** that separates the second flow passage **41B** and the third flow passage **41C**. Further, the second imaginary plane B2 is

located at an end of the wall **44** toward the second connection passage **41E**. The portion extending from the second imaginary plane **B2** in the first negative direction **X2** corresponds to the second connection passage **41E**.

(Noise Cancellation Chamber of Silencer)

[0042] As shown in FIG. **4**, the silencer **40** of the CPAP device **10** includes a first noise cancellation chamber **42A** and a second noise cancellation chamber **42B**. The first noise cancellation chamber **42A** and the second noise cancellation chamber **42B** allow for transmission of pressure fluctuations of the air in the flow passage **41**. The first noise cancellation chamber **42A** and the second noise cancellation chamber **42B** are defined by the walls **44**.

[0043] The first noise cancellation chamber **42A** is located between the second flow passage **41B** and the third flow passage **41C** in a direction parallel to the second axis **Y**. The first noise cancellation chamber **42A** is connected to the second connection passage **41E** through a first opening **45A** included in the wall **44** that separates the first noise cancellation chamber **42A** and the second connection passage **41E**. Further, an open area of the first opening **45A** is greater than a cross-sectional flow area of the second connection passage **41E** at a portion connected to the first opening **45A**. The cross-sectional flow area of the second connection passage **41E** refers to a cross-sectional area of the second connection passage **41E** in a direction orthogonal to the center axis of the second connection passage **41E**.

[0044] The first noise cancellation chamber **42A** includes an inner surface on which an opposing surface **OS** is defined. The opposing surface **OS** opposes the first opening **45A** and is non-perpendicular with respect to a direction in which the first opening **45A** is oriented. Specifically, the opposing surface **OS** is a flat surface inclined with respect to the direction in which the first opening **45A** is oriented. The direction in which the first opening **45A** is oriented coincides with a direction in which the inner surface of the first noise cancellation chamber **42A** is viewed such that the apparent area of an opening surface of the first opening **45A** is maximal. In the present embodiment, the direction in which the first opening **45A** is oriented coincides with the first positive direction **X1**. In other words, the direction in which the first opening **45A** is oriented coincides with a direction orthogonal to the opening surface of the first opening **45A**. When the wall **44** that separates the first noise cancellation chamber **42A** and the second connection passage **41E** is orthogonally projected by a light emitted in the direction in which the first opening **45A** is oriented, the portion of the inner surface of the first noise cancellation chamber **42A** irradiated with the light corresponds to the opposing surface **OS** opposing the first opening **45A**.

[0045] A maximum dimension of the first noise cancellation chamber **42A** in a direction parallel to the first axis **X** is referred to as a first noise cancellation chamber length **SL**, and a maximum dimension of the first noise cancellation chamber **42A** in a direction parallel to the second axis **Y** is referred to as a first noise cancellation chamber width **SW**. A maximum dimension of the second connection passage **41E** in a direction parallel to the first axis **X** is referred to as a second connection passage width **41EW**. A maximum dimension of the second flow passage **41B** in a direction parallel to the second axis **Y** is referred to as a second flow passage width **41BW**. A maximum dimension of the third flow passage **41C** in a direction parallel to the second axis **Y** is referred to as a third flow passage width **41CW**. In this case, the first noise cancellation chamber length **SL** is greater than the second connection passage width **41EW**. The first noise cancellation chamber width **SW** is greater than each of the second flow passage width **41BW** and the third flow passage width **41CW**. A maximum dimension of the first opening **45A** in a direction parallel to the second axis **Y** is referred to as a first opening width **45AW**. The first opening width **45AW** is less than the first noise cancellation chamber width **SW**.

[0046] The second noise cancellation chamber **42B** is located between the second flow passage **41B** and the third flow passage **41C** in a direction parallel to the second axis **Y**. The second noise cancellation chamber **42B** is connected to the third flow passage **41C** through a second opening **45B** included in the wall **44** that separates the second noise cancellation chamber **42B** and the third

flow passage **41C**. A maximum dimension of the second opening **45B** in a direction parallel to the first axis X is less than a maximum dimension of the second noise cancellation chamber **42B** in a direction parallel to the first axis X. Further, an open area of the second opening **45B** is greater than a cross-sectional flow area of the third flow passage **41C**. The cross-sectional flow area of the third flow passage **41C** is defined in the same manner as that of the second connection passage **41E**.

[0047] The second noise cancellation chamber **42B** includes an inner surface on which an opposing surface is defined. The opposing surface opposes the second opening **45B** and is non-perpendicular with respect to a direction in which the second opening **45B** is oriented. Specifically, the opposing surface on the inner surface of the second noise cancellation chamber **42B** is a flat surface inclined with respect to the direction in which the second opening **45B** is directed.

[0048] The maximum dimension of the second noise cancellation chamber **42B** in a direction parallel to the first axis X is greater than the second connection passage width **41EW**. Further, a maximum dimension of the second noise cancellation chamber **42B** in a direction parallel to the second axis Y is greater than each of the second flow passage width **41BW** and the third flow passage width **41CW**. In the present embodiment, the maximum dimension of the second noise cancellation chamber **42B** in a direction parallel to the second axis Y is equal to the first noise cancellation chamber width SW.

[0049] The silencer **40** of the CPAP device **10** includes a sound absorbing material **43** arranged in the first noise cancellation chamber **42A** and the second noise cancellation chamber **42B**. The sound absorbing material **43** is shaped in correspondence with the noise cancellation chambers. That is, each noise cancellation chamber is substantially filled with the sound absorbing material **43**. The sound absorbing material **43** is a porous material. Specifically, the sound absorbing material **43** includes, for example, urethane foam.

[0050] The silencer **40** of the CPAP device **10** includes a film F. The film F closes the first opening **45A**. The film F is thinner than the wall **44** that separates the first noise cancellation chamber **42A** and the second connection passage **41E**. The material of the film F is soft and has a lower elastic coefficient than the wall **44**. The material of the film F includes a synthetic resin, such as polyethylene, polyethylene terephthalate, or the like. Further, another film that is not the film F, which closes the first opening **45A**, closes the second opening **45B**. The film is thinner than the wall **44** that separates the second noise cancellation chamber **42B** and the third flow passage **41C**. The material of the film is the same as that of the film F, which closes the first opening **45A**.

Advantages of the Present Embodiment

[0051] (1) In the above embodiment, the first connection passage length **41DL**, which is the maximum dimension of the first connection passage **41D** in a direction parallel to the second axis Y, differs from the second connection passage length **41EL**, which is the maximum dimension of the second connection passage **41E** in a direction parallel to the second axis Y. Such a difference in the dimension of the connection passages allows for cancellation of noise in different frequency bands in the connection passages. Consequently, noise in various frequency bands is reduced.

[0052] Further, the silencer **40** of the CPAP device **10** includes the first noise cancellation chamber **42A**. The first noise cancellation chamber **42A** is connected to the second connection passage **41E** through the first opening **45A** included in the wall **44** that separates the first noise cancellation chamber **42A** and the second connection passage **41E**. The opening continuous with the first noise cancellation chamber **42A** is arranged at a portion where the flow passage **41** is bent. This facilitates propagation of pressure fluctuations of air, or noise, into the first noise cancellation chamber **42A**. Consequently, noise is effectively cancelled in the first noise cancellation chamber **42A**.

[0053] (2) In the above embodiment, the silencer **40** of the CPAP device **10** further includes the sound absorbing material **43** arranged in the first noise cancellation chamber **42A**. This readily reduces noise propagated into the first noise cancellation chamber **42A**. That is, noise is cancelled more effectively. Further, the sound absorbing material **43** hinders air from flowing into the first

noise cancellation chamber **42A**, thereby ensuring the airflow through the flow passage **41** is unobstructed. This limits the pressure loss of the air flowing through the flow passage **41**. The same applies to the second noise cancellation chamber **42B**.

[0054] (3) In the above embodiment, the opposing surface OS is non-perpendicular with respect to the direction in which the first opening **45A** is oriented. Therefore, the opposing surface OS reflects pressure fluctuations of the air transmitted through the first opening **45A** toward a location other than the first opening **45A**. In other words, the reflected noise will not leak out of the first noise cancellation chamber **42A** through the first opening **45A**. The same applies to the second noise cancellation chamber **42B** and the second opening **45B**.

[0055] (4) In the above embodiment, the first noise cancellation chamber width SW and the first noise cancellation chamber length SL are both greater than each of the second flow passage width **41BW**, the third flow passage width **41CW**, and the second connection passage width **41EW**. That is, the first noise cancellation chamber **42A** has a sufficient volume. Accordingly, noise propagated into the first noise cancellation chamber **42A** will remain in the first noise cancellation chamber **42A** for a relatively long period. This readily reduces the noise, thereby improving the noise cancellation effect. The same applies to the second noise cancellation chamber **42B**.

[0056] (5) In the above embodiment, the second connection passage **41E** is located closer to the blower **20** than the first connection passage **41D** is. In other words, the first noise cancellation chamber **42A** is located near the blower **20**, which is the source of noise. This facilitates propagation of noise into the first noise cancellation chamber **42A**, thereby improving the noise cancellation effect.

[0057] (6) In the above embodiment, the open area of the first opening **45A** is greater than the cross-sectional flow area of the second connection passage **41E**. This increases a ratio of noise propagated into the first noise cancellation chamber **42A** to noise moving straight-ahead in the flow passage **41**, thereby improving the noise cancellation effect. The same applies to the second opening **45B** of the second noise cancellation chamber **42B**.

[0058] (7) In the above embodiment, the first opening width **45AW** is less than the first noise cancellation chamber width SW. Accordingly, even when noise propagated into the first noise cancellation chamber **42A** is reflected by the inner surface of the wall **44** of the first noise cancellation chamber **42A**, the noise will not leak out of the first opening **45A**. This improves the noise cancellation effect. The same applies to the maximum dimension of the second opening **45B** in a direction parallel to the first axis X and the maximum dimension of the second noise cancellation chamber **42B** in a direction parallel to the first axis X.

[0059] (8) In the above embodiment, the first noise cancellation chamber **42A** includes the film F in the first opening **45A**. The film F is thinner than the wall **44** that separates the first noise cancellation chamber **42A** and the second connection passage **41E**. Since the film F is thinner than the wall **44**, noise is propagated into the first noise cancellation chamber **42A**. On the other hand, air flowing through the flow passage **41** will not enter the first noise cancellation chamber **42A**. This limits the pressure loss of the air flowing through the flow passage **41** without adversely affecting the noise cancellation effect. Furthermore, the sound absorbing material **43** arranged in the first noise cancellation chamber **42A** becomes less likely to contact air, so that collection of odor on the sound absorbing material **43** is limited. The same applies to the second noise cancellation chamber **42B**.

[0060] (9) In the above embodiment, the silencer **40** of the CPAP device **10** includes two noise cancellation chambers, namely, the first noise cancellation chamber **42A** and the second noise cancellation chamber **42B**. These two noise cancellation chambers reduce noise in a relatively wide frequency band.

[0061] (10) In the above embodiment, pressure fluctuations of the air in the second connection passage **41E** are readily transmitted into the first noise cancellation chamber **42A**, and pressure fluctuations of the air in the third flow passage **41C** are readily transmitted to the second noise

cancellation chamber **42B**. In this manner, noise is reduced at different locations, thereby improving the noise cancellation effect.

[0062] (11) In the above embodiment, the second noise cancellation chamber **42B** is located closer to the blower **20**, or the source of noise, than the first noise cancellation chamber **42A** is. Therefore, noise is reduced efficiently.

Modified Examples

[0063] The above embodiment and the following modifications may be combined as long as the combined modifications remain technically consistent with each other.

[0064] The main unit MU and the sub-unit SU may have any shape. The shapes of the units may be freely designed as long as the functionality of the CPAP device **10** is implemented.

[0065] The sub-unit SU does not have to include the humidifier **30**.

[0066] The CPAP device **10** may include another unit in addition to the main unit MU and the sub-unit SU. Alternatively, the CPAP device **10** may be formed as a single unit that incorporates the main unit MU and the sub-unit SU.

[0067] The silencer **40** of the CPAP device **10** may be located at a downstream side of the blower **20**. For example, the downstream passage **62** of the main unit MU may be connected to the first inflow passage **51** of the sub-unit SU.

[0068] The power supply **101** is not limited to the example described in the above embodiment, as long as the power supply **101** is configured to supply power to the controller **100** and the blower **20**. For example, the main unit MU may include a terminal connected to an AC adapter, and the controller **100** and the blower **20** may be supplied with power from a commercial power supply via the AC adapter.

[0069] The direction in which the first flow passage **41A** is arranged next to the second flow passage **41B** does not have to be the same as the direction in which the second flow passage **41B** is arranged next to the third flow passage **41C**. For example, the first flow passage **41A** may be arranged next to the second flow passage **41B** in a direction parallel to the second axis Y, and the second flow passage **41B** may be arranged next to the third flow passage **41C** in a direction parallel to the third axis Z.

[0070] There may be four or more flow passages that extend along the first axis X. In this case, for example, a third connection passage connects the third flow passage **41C** and a fourth flow passage.

[0071] In the above embodiment, the first flow passage **41A**, the second flow passage **41B**, and the third flow passage **41C** are specified in this order from the upstream side for illustrative purposes. However, any passage may be specified as “second flow passage” or “third flow passage”. In other words, as long as the first noise cancellation chamber **42A** is formed between any two of the three passages in the above embodiment, it can be considered that the first noise cancellation chamber **42A** is defined between “second flow passage” and “third flow passage”.

[0072] The shape of the flow passage **41** is not limited to the example described in the above embodiment. For example, the walls **44** defining the first flow passage **41A**, the second flow passage **41B**, and the third flow passage **41C** do not have to be straight. For example, the walls **44** may be undulated or have a plurality of protrusions as in a suppressor. Furthermore, inner surfaces of the walls **44** of the flow passage **41** may be covered with the sound absorbing material **43**.

[0073] The first opening width **45AW** may be equal to the first noise cancellation chamber width SW. Even in this case, advantage (1) is obtained. Furthermore, the maximum dimension of the second opening **45B** in a direction parallel to the first axis X may be equal to the maximum dimension of the second noise cancellation chamber **42B** in a direction parallel to the first axis X. Even in this case, advantages (1) and (9) are obtained.

[0074] The open area of the first opening **45A** may be less than or equal to the cross-sectional flow area of the second connection passage **41E**. Even if the open area of the first opening **45A** is relatively small, advantage (1) will be obtained as long as pressure fluctuations of the air in the

flow passage **41** can be transmitted through the first opening **45A**. Furthermore, the open area of the second opening **45B** may be less than or equal to the cross-sectional flow area of the third flow passage **41C**. For the same reason, advantages (1) and (9) are obtained.

[0075] The opposing surface OS on the inner surface of the first noise cancellation chamber **42A**, which opposes the first opening **45A**, may be perpendicular with respect to the direction in which the first opening **45A** is oriented. Even in this case, advantage (1) is obtained. Furthermore, the opposing surface on the inner surface of the second noise cancellation chamber **42B**, which opposes the second opening **45B**, may be perpendicular with respect to the direction in which the second opening **45B** is oriented. Even in this case, advantages (1) and (9) are obtained.

[0076] The first noise cancellation chamber length SL may be less than or equal to the second connection passage width **41EW**. Furthermore, the first noise cancellation chamber width SW may be less than or equal to the second flow passage width **41BW**, and the first noise cancellation chamber width SW may be less than or equal to the third flow passage width **41CW**. Even if the first noise cancellation chamber **42A** is relatively small, advantage (1) is obtained.

[0077] The maximum dimension of the second noise cancellation chamber **42B** in a direction parallel to the first axis X may be less than or equal to the second connection passage width **41EW**. Furthermore, the maximum dimension of the second noise cancellation chamber **42B** in a direction parallel to the second axis Y may be less than or equal to the second flow passage width **41BW**, and the maximum dimension of the second noise cancellation chamber **42B** in a direction parallel to the second axis Y may be less than or equal to the third flow passage width **41CW**. Even if the second noise cancellation chamber **42B** is relatively small, advantages (1) and (9) are obtained.

[0078] The silencer **40** of the CPAP device **10** does not have to include the second noise cancellation chamber **42B**. As long as the first noise cancellation chamber **42A** is arranged, advantage (1) is obtained.

[0079] The silencer **40** of the CPAP device **10** may have three or more noise cancellation chambers. Furthermore, the silencer **40** of the CPAP device **10** may include a noise cancellation chamber between the first flow passage **41A** and the second flow passage **41B** in a direction parallel to the second axis Y. In this case, the wall **44** that separates the noise cancellation chamber and the flow passage **41** may include any number of openings at any locations. Preferably, an opening is included in the wall **44** that separates the noise cancellation chamber and the first connection passage **41D**.

[0080] The second opening **45B** may be included in the wall **44** that separates the second noise cancellation chamber **42B** and the second flow passage **41B**, instead of the wall **44** that separates the second noise cancellation chamber **42B** and the third flow passage **41C**. Alternatively, the second opening **45B** may be included in the wall **44** that separates the second noise cancellation chamber **42B** and the third flow passage **41C**, and another opening may be included in the wall **44** that separates the second noise cancellation chamber **42B** and the second flow passage **41B**. Preferably, an opening is arranged in at least one of the wall **44** that separates the second noise cancellation chamber **42B** and the second flow passage **41B** and the wall **44** that separates the second noise cancellation chamber **42B** and the third flow passage **41C**.

[0081] The third flow passage **41C** may be located farther from the blower **20** than the second flow passage **41B** is with respect to the airflow direction. Specifically, for example, the following structure may be adopted. One end of the third flow passage **41C** is connected to the second end of the first inflow passage **51**. The second flow passage **41B** is located at a side of the third flow passage **41C** in the second positive direction Y1, and the first flow passage **41A** is located at a side of the second flow passage **41B** in the second positive direction Y1. One end of the first flow passage **41A** is connected to the first end of the first outflow passage **52**. The second connection passage **41E** connects the other end of the third flow passage **41C** and one end of the second flow passage **41B**. The first connection passage **41D** connects the other end of the second connection passage **41E** and the other end of the first flow passage **41A**.

[0082] Even with such a structure, in which the second connection passage **41E** is located farther from the blower **20** than the first connection passage **41D** is with respect to the airflow direction, advantage (1) is obtained.

[0083] Also, even with this structure, in which the third flow passage **41C** is located farther from the blower **20** than the second flow passage **41B** is with respect to the airflow direction, and the wall **44** that separates the second noise cancellation chamber **42B** and the third flow passage **41C** includes the second opening **45B**, advantages (1) and (9) are obtained.

[0084] The silencer **40** of the CPAP device **10** does not have to include the sound absorbing material **43** arranged in the first noise cancellation chamber **42A**. Even if the sound absorbing material **43** is not arranged, the noise cancellation effect is improved by the shape of the flow passage **41** and formation of the first noise cancellation chamber **42A**. The same applies to the second noise cancellation chamber **42B**.

[0085] The material of the sound absorbing material **43** is not limited to the example described in the above embodiment. For example, the material of the sound absorbing material **43** may include polyester, glass wool, felt, or the like. The sound absorbing material **43** may be any porous material.

[0086] The first opening **45A** does not have to be closed by the film F. Even in this case, advantage (1) is obtained. Furthermore, the second opening **45B** does not have to be closed by the film F. Even in this case, advantages (1) and (9) are obtained.

[0087] The material of the film F is not limited to the example described in the above embodiment. For example, the film F may be made of paper.

Clauses

[0088] Technical concepts obtained from the above embodiment and the modifications will now be described.

[0089] (1) A silencer for a CPAP device, the silencer including: a flow passage through which air flows in accordance with operation of a blower; and a noise cancellation chamber that allows for transmission of pressure fluctuations of the air in the flow passage, in which: the flow passage includes a first flow passage extending along a first axis, a second flow passage extending along the first axis and arranged next to the first flow passage in a direction orthogonal to the first axis, a third flow passage extending along the first axis and arranged next to the second flow passage in the direction orthogonal to the first axis, a first connection passage connecting an end of the first flow passage in a positive direction and an end of the second flow passage in the positive direction, the positive direction being one of two directions parallel to the first axis, and a second connection passage connecting an end of the second flow passage in a negative direction and an end of the third flow passage in the negative direction, the negative direction being another one of the two directions parallel to the first axis and being opposite to the positive direction; a maximum dimension of the first connection passage in a direction in which the second flow passage is arranged next to the third flow passage differs from a maximum dimension of the second connection passage in the direction in which the second flow passage is arranged next to the third flow passage; the noise cancellation chamber is located between the second flow passage and the third flow passage in the direction in which the second flow passage is arranged next to the third flow passage; and the noise cancellation chamber is connected to the second connection passage through an opening included in a wall that separates the noise cancellation chamber and the second connection passage.

[0090] (2) The silencer according to clause (1), further including a sound absorbing material arranged in the noise cancellation chamber.

[0091] (3) The silencer according to clause (1) or (2), in which the noise cancellation chamber includes an inner surface on which an opposing surface is defined, the opposing surface opposing an opening surface of the opening and being non-perpendicular with respect to a direction orthogonal to the opening surface.

[0092] (4) The silencer according to any one of clauses (1) to (3), in which a maximum dimension of the noise cancellation chamber in a direction parallel to the first axis is greater than a maximum dimension of the second connection passage in the direction parallel to the first axis, and a maximum dimension of the noise cancellation chamber in the direction in which the second flow passage is arranged next to the third flow passage is greater than each of a maximum dimension of the second flow passage in the direction in which the second flow passage is arranged next to the third flow passage and a maximum dimension of the third flow passage in the direction in which the second flow passage is arranged next to the third flow passage.

[0093] (5) The silencer according to any one of clauses (1) to (4), in which the second connection passage is located closer to the blower than the first connection passage is with respect to an airflow direction.

[0094] (6) The silencer according to any one of clauses (1) to (5), in which an open area of the opening is greater than a cross-sectional flow area of the second connection passage.

[0095] (7) The silencer according to any one of clauses (1) to (6), in which a maximum dimension of the opening in the direction in which the second flow passage is arranged next to the third flow passage is less than a maximum dimension of the noise cancellation chamber in the direction in which the second flow passage is arranged next to the third flow passage.

[0096] (8) The silencer according to any one of clauses (1) to (7), further including: a film thinner than the wall that separates the noise cancellation chamber and the second connection passage, in which the film closes the opening.

[0097] (9) The silencer according to any one of clauses (1) to (8), further including: when the noise cancellation chamber is a first noise cancellation chamber, and the opening that connects the first noise cancellation chamber and the second connection passage is a first opening, a second noise cancellation chamber that allows for transmission of pressure fluctuations of the air in the flow passage, in which the second noise cancellation chamber is located between the second flow passage and the third flow passage in the direction in which the second flow passage is arranged next to the third flow passage, and the second noise cancellation chamber is connected to the flow passage through a second opening included in a wall that separates the second noise cancellation chamber and the flow passage.

[0098] (10) The silencer according to clause (9), in which the second opening is included in a wall that separates the second noise cancellation chamber and the second flow passage or a wall that separates the second noise cancellation chamber and the third flow passage.

[0099] (11) The silencer according to clause (10), in which the third flow passage is located closer to the blower than the second flow passage is with respect to an airflow direction, and the wall that separates the second noise cancellation chamber and the third flow passage includes the second opening.

[0100] (12) A CPAP device, including: a blower that blows air; and a silencer, in which: the silencer includes a flow passage through which air flows in accordance with operation of the blower, and a noise cancellation chamber that allows for transmission of pressure fluctuations of the air in the flow passage; the flow passage includes a first flow passage extending along a first axis, a second flow passage extending along the first axis and arranged next to the first flow passage in a direction orthogonal to the first axis, a third flow passage extending along the first axis and arranged next to the second flow passage in the direction orthogonal to the first axis, a first connection passage connecting an end of the first flow passage in a positive direction and an end of the second flow passage in the positive direction, the positive direction being one of two directions parallel to the first axis, and a second connection passage connecting an end of the second flow passage in a negative direction and an end of the third flow passage in the negative direction, the negative direction being another one of the two directions parallel to the first axis and being opposite to the positive direction; a maximum dimension of the first connection passage in a direction in which the second flow passage is arranged next to the third flow passage differs from a

maximum dimension of the second connection passage in the direction in which the second flow passage is arranged next to the third flow passage; the noise cancellation chamber is located between the second flow passage and the third flow passage in the direction in which the second flow passage is arranged next to the third flow passage; and the noise cancellation chamber is connected to the second connection passage through an opening included in a wall that separates the noise cancellation chamber and the second connection passage.

[0101] (13) The CPAP device according to clause (12), in which the second connection passage is located closer to the blower than the first connection passage is with respect to an airflow direction.

[0102] (14) The CPAP device according to clause (12) or (13), further including: when the noise cancellation chamber is a first noise cancellation chamber, and the opening that connects the first noise cancellation chamber and the second connection passage is a first opening, a second noise cancellation chamber that allows for transmission of pressure fluctuations of the air in the flow passage, in which the second noise cancellation chamber is located between the second flow passage and the third flow passage in the direction in which the second flow passage is arranged next to the third flow passage, the second noise cancellation chamber is connected to the flow passage through a second opening included in a wall that separates the second noise cancellation chamber and the third flow passage, and the third flow passage is located closer to the blower than the second flow passage is with respect to the airflow direction. [0103] **10)** CPAP device [0104] MU) main unit [0105] SU) sub-unit [0106] **40)** silencer [0107] **44)** wall [0108] **41)** flow passage [0109] **41A)** first flow passage [0110] **41B)** second flow passage [0111] **41BW)** second flow passage width [0112] **41C)** third flow passage [0113] **41CW)** third flow passage width [0114] **41D)** first connection passage [0115] **41DL)** first connection passage length [0116] **41E)** second connection passage [0117] **41EL)** second connection passage length [0118] **41EW)** second connection passage width [0119] **42A)** first noise cancellation chamber [0120] SW) first noise cancellation chamber width [0121] SL) first noise cancellation chamber length [0122] **42B)** second noise cancellation chamber [0123] **45A)** first opening [0124] **45AW)** first opening width [0125] **45B)** second opening [0126] **43)** noise absorbing material [0127] OS) opposing surface [0128] F) film

Claims

1. A silencer for a CPAP device, the silencer comprising: a flow passage through which air flows in accordance with operation of a blower; and a noise cancellation chamber configured to allow for transmission of pressure fluctuations of the air in the flow passage, wherein: the flow passage includes a first flow passage extending along a first axis, a second flow passage extending along the first axis and arranged next to the first flow passage in a direction orthogonal to the first axis, a third flow passage extending along the first axis and arranged next to the second flow passage in the direction orthogonal to the first axis, a first connection passage connecting an end of the first flow passage in a positive direction and an end of the second flow passage in the positive direction, the positive direction being one of two directions parallel to the first axis, and a second connection passage connecting an end of the second flow passage in a negative direction and an end of the third flow passage in the negative direction, the negative direction being another one of the two directions parallel to the first axis and being opposite to the positive direction; a maximum dimension of the first connection passage in a direction in which the second flow passage is arranged next to the third flow passage differs from a maximum dimension of the second connection passage in the direction in which the second flow passage is arranged next to the third flow passage; the noise cancellation chamber is located between the second flow passage and the third flow passage in the direction in which the second flow passage is arranged next to the third flow passage; and the noise cancellation chamber is connected to the second connection passage through an opening included in a wall separating the noise cancellation chamber and the second

connection passage.

2. The silencer according to claim 1, further comprising a sound absorbing material arranged in the noise cancellation chamber.

3. The silencer according to claim 1, wherein the noise cancellation chamber includes an inner surface on which an opposing surface is defined, the opposing surface opposing an opening surface of the opening and being non-perpendicular with respect to a direction orthogonal to the opening surface.

4. The silencer according to claim 1, wherein a maximum dimension of the noise cancellation chamber in a direction parallel to the first axis is greater than a maximum dimension of the second connection passage in the direction parallel to the first axis, and a maximum dimension of the noise cancellation chamber in the direction in which the second flow passage is arranged next to the third flow passage is greater than each of a maximum dimension of the second flow passage in the direction in which the second flow passage is arranged next to the third flow passage and a maximum dimension of the third flow passage in the direction in which the second flow passage is arranged next to the third flow passage.

5. The silencer according to claim 1, wherein the second connection passage is located closer to the blower than the first connection passage is with respect to an airflow direction.

6. The silencer according to claim 1, wherein an open area of the opening is greater than a cross-sectional flow area of the second connection passage.

7. The silencer according to claim 1, wherein a maximum dimension of the opening in the direction in which the second flow passage is arranged next to the third flow passage is less than a maximum dimension of the noise cancellation chamber in the direction in which the second flow passage is arranged next to the third flow passage.

8. The silencer according to claim 1, further comprising: a film thinner than the wall separating the noise cancellation chamber and the second connection passage, wherein the film closes the opening.

9. The silencer according to claim 1, further comprising: when the noise cancellation chamber is a first noise cancellation chamber, and the opening connecting the first noise cancellation chamber and the second connection passage is a first opening, a second noise cancellation chamber configured to allow for transmission of pressure fluctuations of the air in the flow passage, wherein the second noise cancellation chamber is located between the second flow passage and the third flow passage in the direction in which the second flow passage is arranged next to the third flow passage, and the second noise cancellation chamber is connected to the flow passage through a second opening included in a wall separating the second noise cancellation chamber and the flow passage.

10. The silencer according to claim 9, wherein the second opening is included in a wall separating the second noise cancellation chamber and the second flow passage or a wall separating the second noise cancellation chamber and the third flow passage.

11. The silencer according to claim 10, wherein the third flow passage is located closer to the blower than the second flow passage is with respect to an airflow direction, and the wall separating the second noise cancellation chamber and the third flow passage includes the second opening.

12. A CPAP device, comprising: a blower configured to blow air; and a silencer, wherein: the silencer includes a flow passage through which air flows in accordance with operation of the blower, and a noise cancellation chamber configured to allow for transmission of pressure fluctuations of the air in the flow passage; the flow passage includes a first flow passage extending along a first axis, a second flow passage extending along the first axis and arranged next to the first flow passage in a direction orthogonal to the first axis, a third flow passage extending along the first axis and arranged next to the second flow passage in the direction orthogonal to the first axis, a first connection passage connecting an end of the first flow passage in a positive direction and an end of the second flow passage in the positive direction, the positive direction being one of two

directions parallel to the first axis, and a second connection passage connecting an end of the second flow passage in a negative direction and an end of the third flow passage in the negative direction, the negative direction being another one of the two directions parallel to the first axis and being opposite to the positive direction; a maximum dimension of the first connection passage in a direction in which the second flow passage is arranged next to the third flow passage differs from a maximum dimension of the second connection passage in the direction in which the second flow passage is arranged next to the third flow passage; the noise cancellation chamber is located between the second flow passage and the third flow passage in the direction in which the second flow passage is arranged next to the third flow passage; and the noise cancellation chamber is connected to the second connection passage through an opening included in a wall separating the noise cancellation chamber and the second connection passage.

13. The CPAP device according to claim 12, wherein the second connection passage is located closer to the blower than the first connection passage is with respect to an airflow direction.

14. The CPAP device according to claim 13, further comprising: when the noise cancellation chamber is a first noise cancellation chamber, and the opening connecting the first noise cancellation chamber and the second connection passage is a first opening, a second noise cancellation chamber configured to allow for transmission of pressure fluctuations of the air in the flow passage, wherein the second noise cancellation chamber is located between the second flow passage and the third flow passage in the direction in which the second flow passage is arranged next to the third flow passage, the second noise cancellation chamber is connected to the flow passage through a second opening included in a wall separating the second noise cancellation chamber and the third flow passage, and the third flow passage is located closer to the blower than the second flow passage is with respect to the airflow direction.

15. The silencer according to claim 2, wherein the noise cancellation chamber includes an inner surface on which an opposing surface is defined, the opposing surface opposing an opening surface of the opening and being non-perpendicular with respect to a direction orthogonal to the opening surface.

16. The silencer according to claim 2, wherein a maximum dimension of the noise cancellation chamber in a direction parallel to the first axis is greater than a maximum dimension of the second connection passage in the direction parallel to the first axis, and a maximum dimension of the noise cancellation chamber in the direction in which the second flow passage is arranged next to the third flow passage is greater than each of a maximum dimension of the second flow passage in the direction in which the second flow passage is arranged next to the third flow passage and a maximum dimension of the third flow passage in the direction in which the second flow passage is arranged next to the third flow passage.

17. The silencer according to claim 3, wherein a maximum dimension of the noise cancellation chamber in a direction parallel to the first axis is greater than a maximum dimension of the second connection passage in the direction parallel to the first axis, and a maximum dimension of the noise cancellation chamber in the direction in which the second flow passage is arranged next to the third flow passage is greater than each of a maximum dimension of the second flow passage in the direction in which the second flow passage is arranged next to the third flow passage and a maximum dimension of the third flow passage in the direction in which the second flow passage is arranged next to the third flow passage.

18. The silencer according to claim 2, wherein the second connection passage is located closer to the blower than the first connection passage is with respect to an airflow direction.

19. The silencer according to claim 3, wherein the second connection passage is located closer to the blower than the first connection passage is with respect to an airflow direction.

20. The silencer according to claim 4, wherein the second connection passage is located closer to the blower than the first connection passage is with respect to an airflow direction.
